Evaluation of Synergy-Based Hand Gesture Recognition Method Against Force Variation for Robust Myoelectric Control

Zhicheng Teng, Guanghua Xu , Member, IEEE, Renghao Liang, Min Li , Member, IEEE, and Sicong Zhang

Abstract— The non-stationary characteristics of surface electromyography (sEMG) and possible adverse variations in real-world conditions make it still an open challenge to realize robust myoelectric control (MEC) for multifunctional prostheses. Variable muscle contraction level is one of the handicaps that may degrade the performance of MEC. In this study, we proposed a force-invariant intent recog- nition method based on muscle synergy analysis (MSA) in the setting of three self-defined force levels (low, medium, and high). Specifically, a fast matrix factorization algo- rithm based on alternating non-negativity constrained least squares (NMF/ANLS) was chosen to extract task-specific synergies associated with each of six hand gestures in the training stage; while for the testing samples, we used the non-negativeleastsquare (NNLS) method to estimate neural commands for movement classification.The performance of proposed method was compared with conventional pattern recognition (PR) method consisting of LDA (linear discrim- ination analysis) classifier and representative features in three offline evaluation scenarios. Statistical tests on ten able-bodied subjects revealed no significant difference in intra-force-level($p = 0.353$) and multi-force-level($p = 0.695$) accuracy; But the synergy-based method performed sig- nificantly better than conventional PR-based method under inter-force-level conditions ($p < 0.05$). Similar results were observedfornine amputee subjectsthough there was a drop in the classificationaccuracy.This study was the first to con- currently demonstrate the robustness and predictive power of task-specific synergies under variant force levels and explore their potential for reliable intent recognition against force variation. Although the online performance is yet to be demonstrated, the proposed method is characterized by simple training procedure and acceptable computational efficiency, which would potentially provide an alternative approach for the developmentof clinicallyviable prostheses and rehabilitation robots driven by sEMG.

Index Terms— sEMG, myoelectric control, multifunctional prostheses, muscle contraction level, muscle synergy.

I. INTRODUCTION P

OWERED myoelectric prostheses can, to some extent, help amputees regain the lost function of their residual limbs and rebuild confidence in daily life. Despite the fact that the mechanical design of state-of-the-art robotic hands enables vivid movements imitation of human hands [1], these dexterous anthropomorphic hands are beyond direct appli- cation in myoelectric prostheses due to the incommensurate performance of available human-machine-interface (HMI).

Over the past decades, pattern recognition (PR) based myoelectric classification scheme has been widely explored for multifunctional prostheses control [2], [3]. Such scheme employs multichannel EMG signals and provides intuitive control with more active degree of freedoms (DOFs). Despite a large amount of promising results (with the average accuracy above 95%) reported in controlled laboratory environments, there exists a noticeable disparity between ideal and real-world conditions. It has been confirmed that the reliability of myoelectric control was highly dependent on physical and physiological constraints, including fixed electrode location on specified muscles, specific limb orientation, steady muscle contraction level, avoidance of muscle fatigue, and so on. These favorable conditions, however, are not likely to meet in a practical context for prostheses control. It is for these reasons that PR-based control scheme meets its Waterloo in clinical translation. To our knowledge, only one spin-off company [4] offers a classification-based controller as an add-on for various prostheses. Consequently, the rejection rate of multifunctional myoelectric prostheses and the use of non-functional cosmetic prostheses remains stay at a high level [5], [6].

Researchers have made multipronged efforts to improve the performance of PR-based control scheme in the presence of variant muscle contraction levels. Tolooshams and Jiang [7] and Pradhan et al. [8] proposed a signal pre-processing method using frequency division technique (FDT) for online myo- electric PR classification. The results indicated that FDT had an advantage over traditional methods in the tested real-time myoelectric control tasks. As for the classifiers, Li et al. [9]

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This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the Ethics Committee of University of Technology Sydney and Plymouth University.

The authors are with the School of Mechanical Engineering, Institute of Engineering and Medicine Interdisciplinary Studies, Xi'an Jiaotong University, Xi'an 710049, China (e-mail: ghxu@xjtu.edu.cn).

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proposed a parallel classification strategy consisting of three classifiers trained with data acquired at different force levels, and designed a regulation to categorize the input signals into corresponding classifiers for robust PR. Moin et al. [10] used a brain-inspired hyperdimensional computing paradigm to build classification models that were both robust to force variations and able to recognize multiple contraction levels. Powar and Chemmangat [11] proposed the dynamic time warping (DTW) method to decrease the effect of force variations on EMG PR, and achieved a significant 10% rise in classification accuracy. Researchers also proposed robust feature extraction (FE) technique against force variation for EMG PR system, and some encouraging results were reported [12]–[15].

One major shortcoming for PR-based classification scheme may rest with its incompetence in making full use of underlying neurophysiological information hidden in sEMG data. In other words, crucial information involved in the processes of neural control to generate observable muscle activation patterns (MAP) cannot be explicitly incorporated into PR-based system [16]. Though including all data at different muscle contraction levels into the training procedure may mitigate such adverse effects, the increased training time and limited performance enhancement is still a problem for the clinical translation of this technology.

Muscle synergies (MS) have been proposed as build- ing blocks that could simplify the construction of complex motor behaviors. This theory implies that central nervous system (CNS) may not directly activate individual muscle one by one to realize motor control. Instead, a group of muscles is activated simultaneously by temporal neural commands from CNS to perform multiple DOF movements in a coordinated manner [17]. Although the existence of MS is still an open question, several advanced HMI systems have been developed using the MS framework. Jiang et al. [18], [19] extracted wrist kinematics from multi-channel EMG using a DOF- wise non-negative matrix factorization algorithm to realize proportional and simultaneous control of two or three DOFs. Ma et al. [20] proposed a control scheme that smooths the synergy-based inputs, thereby stabilizing the movement control. Ison et al. [21], [22] demonstrated simultaneous and proportional control of a 4-DOF myoelectric interface by using muscle synergy-inspired inputs from high-density EMG. However, all these studies provided reliable control at the expense of limited DOF. Researchers also explored the role of MS in multi-DOF movement classification. However, they either used the extracted muscle synergies or the activation coefficients as the feature vector to train a classifier for task discrimination, which was essentially still a conventional PR system and the predictive power of MS was not fully investigated [23]–[26]. Ajiboye and Weir [27] verified the predictive framework of MS by extracting eight synergies from eleven postures to allow for at least 90% prediction of the EMG patterns of all 33 postures, which throw light on synergy-based movement classification. Rasool et al. [16] then presented a novel formulation that employs task-specific synergies and state-space representation of neural signals to discriminate six hand movements. Extending Rasool's work, Zhang et al. [28] built a simultaneous control scheme by employing one synergy-based classifier for each finger gesture. However, none of the above work investigated the role of MS in developing force-invariant classification scheme. Patel et al. [29] proposed a classification method against force variation inspired by muscle coordination. While informative, they failed to utilize a well-constructed synergy model for muscle synergy analysis. Taborri et al. [30] evaluated the robustness of EMG reconstruction via MS and the repeatability of MS parameters as potential neurophysiological indices. Deng et al. [31] and Geng et al. [32] clarified the modulation of MS for multiple forearm movements under variant force levels and arm positions. Nevertheless, they did not propose a classification scheme utilizing MS to realize robust movement classification.

In the present study, we concurrently demonstrated the effect of variant muscle contraction levels on both the constitution of estimated MS and the performance of a synergy-based classification method. Firstly, a fast matrix factorization algorithm based on alternating non-negativity constrained least squares (NMF/ANLS) was used to extract task-specific synergies for six hand gestures. Then the structure similarity and predictive power of estimated synergies were investigated in the setting of three self-defined muscle con- traction levels. Following that, a novel classification scheme against force variation based on muscle synergy analysis was proposed accordingly. Finally, the performance of proposed algorithm was evaluated and compared with conventional PR-based method consisting of linear discrimination analy- sis (LDA) classifier and representative features reported in literature. The outcome of this study would benefit the design of clinically viable prosthetic hand.

II. MATERIALS AND METHODS

A. sEMG Data Collection and Preprocessing

We use EMG data from two publicly available datasets to demonstrate the effectiveness of proposed classification scheme. Due to the length of the paper, only the first dataset was used to illustrate the robustness and predictive power of muscle synergies. A brief description of these two datasets is presented here for the structural integrity of this paper.

1) Able-Bodied Dataset: The first dataset from Khushaba et al. [33] involves EMG signals recorded from ten able-bodied subjects. Their age varied from 20 to 30, with forearm diameter of 26.6 ± 2.4 cm. Participants had no previous experience with EMG-controlled HMI. They were given a 20-min briefing session to get familiar with the experimental protocol. All procedures were approved by the research ethics committee at University of Technology Sydney.

During the experiment, subjects sat comfortably in front of a standard computer screen. Instructions appeared clearly on the computer screen using an image of the desired movement along with a text caption. The subjects performed six movements including hand close (HC), hand open (HO), wrist extension (WE), wrist flexion (WF), wrist ulnar deviation (WU), and wrist radial deviation (WR). Each of these six movements was repeated three times at three self-defined muscle contraction levels, i.e., low (L), medium (M), and high (H). A virtual instrument implemented in LABVIEW

(National Instruments, USA) was used for signal acquisition and display, which helps the subjects produce the needed force level. The subjects were asked to hold each gesture for a duration of five seconds in each trial. There is a 10 s rest between trials to avoid muscle fatigue. The sEMG signals were collected at the sampling rate of 4000 Hz, with six electrodes (Bagnoli desktop EMG system, Delsys Inc., USA) equally spaced across the circumference of the forearm for each subject. This dataset also contains wrist accelerometer data, which was not included in the present study. For more information about the metadata, please refer to [33].

2) Amputee Dataset: The second dataset from Al-Timemy et al. [12] consists of EMG signals recorded from nine transradial amputees (seven traumatic and two congenital). All subjects missed their left hand except one with right hand amputation. The first seven subjects had never used myoelectric prostheses while the remaining two used one for a certain time. The local ethical committee at the School of Computing and Mathematics, Plymouth University approved this research.

Six movements were considered in the experiment, includ- ing thumb flexion, index flexion, fine pinch, tripod grip, hook grip, and spherical grip. Each movement was performed at three self-defined muscle contraction levels, i.e., low, medium, and high. For each force level, five to eight trials were recorded for each amputee where each trial is hold for a duration of 8-12 s. The amputees used their intact-hand to help them to imagine the needed movement with the required force. In addition, they used visual feedback from the LABVIEW screen to see how the EMG amplitude changes with the force, which was useful for them to produce the needed force level. EMG signals were collected at a sampling rate of 2000 Hz, with eight pairs of Ag/AgCl electrodes (Tyco healthcare, Germany) placed around the stump for each amputee. Data from the first three trials was used in the present study. For detailed description of the dataset please refer to [12].

3) Preprocessing: The recorded raw sEMG signals from the datasets for each movement were then band-pass filtered (4th order Butterworth filter, 20-450 Hz), full-wave rectified, and low-pass filtered (4th order Butterworth filter, 20 Hz) to obtain the envelope of the sEMG signals.

B. Muscle Synergy Model

The time-invariant muscle synergy model indicates that the high-dimensional MAP can be represented as linear combination of low-dimensional muscle synergies that are activated by corresponding activation coefficients [17]

$$m(t) = \sum_{i=1}^{k} h_i(t)\, w_i, \quad (1)$$

where m(t) is a vector of MAP for a group of muscles at time t, each wi, namely the muscle synergy, is a vector representing a fixed weighting of activation within the muscles, each hi(t) is the time-varying combination coefficients hypothesized as the neural commands for the activation of wi, k is the number of muscle synergies. Equation (1) can be written as matrix form with regard to n time instances over a specific time scale:

$$M_{m\times n} = W_{m\times k} H_{k\times n} = [w_1\ w_2 \cdots w_k] \begin{bmatrix} h_1 \\ h_2 \\ \cdots \\ h_k \end{bmatrix}, (k < m) \quad (2)$$

where M is the MAP matrix of m muscles, W is the muscle synergy matrix, H is the activation coefficient matrix. Please note that M is known whereas W and H is unknown in this case.

C. Muscle Activation Patterns Approximation

The recorded sEMG data was processed and divided into several blocks, with each block corresponding to a specific hand gesture. Data from each block were then segmented into 250 ms analysis window with an increment of 30 ms, and features were extracted from each analysis window to constitute the feature matrix M for each hand gesture. This matrix was referred to as the MAP matrix. The resulting samples were then normalized for each subject and each muscle to the maximum activation in that muscle over all trials of all tasks.

The most commonly used method for MAP approxima- tion in muscle synergy analysis (MSA) is the EMG linear envelopes, researchers also used typical time domain (TD) features, i.e., root mean square (RMS) and mean absolute value (MAV) to produce MAP matrix. Ma et al. [20] evaluated the performance of a set of TD features in MSA by comparing the goodness of reconstruction using extracted MS, and find that the best three features are MAV, RMS and waveform length (WL). In this study, however, we put more focus on the influence of different features on classification accuracy and seeking to find the best MAP approximation method for synergy-based movement recognition. The features evaluated in this study were RMS, MAV, and a feature based on discrete Fourier transform and root (DFTR) [13].

D. Extraction of Muscle Synergies

1) Matrix Factorization Method: Muscle synergies can then be estimated from the MAP matrix using factorization algorithms, i.e., non-negative matrix factorization (NMF) [34], independent component analysis (ICA) [35], and principle component analysis (PCA) [36]. NMF is widely used since it constrains the estimated synergies to be no-negative, and thus allowing for more physiological interpretation of the results (muscles activity and neural control signals are inherently no-negative). In contrast, The results of PCA and ICA are either orthogonal or statistically independent. We accordingly chose a NMF algorithm based on alternating non-negativity constrained least squares (NMF/ANLS) that showed fast overall convergence and generated more physically reliable solutions [37].

It should be noted that the NMF/ANLS algorithm usually generates multiple solutions for Equation (2) once the synergy number k is determined. For this reason, the algorithm was

replicated 10 times in this study to extract the set of muscle synergies with the least error for data reconstruction. We used the coefficient of determination (R2) [38] as a criteria to measure the goodness of reconstruction for original sEMG data by estimated muscle synergies and corresponding activation coefficients:

$$R2 = 1 - \frac{\|M - WH\|_2}{\|M - \bar{M}\|_2} \quad (3)$$

where $\|\cdot\|_2$ denotes the squared Frobenius norm of a matrix, and $\bar{M}$ is the mean of the MAP matrix.

We also computed the variance accounted for (VAF), which is a similar criterion calculated on uncentered data:

$$VAF = 1 - \frac{\|M - WH\|_2}{\|M\|_2} \quad (4)$$

2) Determination of Synergy Number: The optimal number of muscle synergies (dimensionality of neural commands from CNS) in the tested model is a hyper-parameter that should be determined before synergy extraction. In general, k is determined as the minimum number of muscle synergies able to explain the structural variation of the dataset, that is, additional synergies only capture noise-generated variation [38]. We plotted the curve of R2 as functions of synergy number k (changing from 1 to 6) to depict the influence of synergy number on the reconstruction goodness of the original data. In this study, the optimal synergy number was determined at the point where R2 > 0.9, namely at least 90 percent of the variance in the data could be explained by the reduced dimension. To avoid data leakage, we only used one out of three EMG trials to separate the data used for determining the number of synergies from the data used for the accuracy testing. Reported for each movement performed by each subject under each force level were the optimal number for estimated muscle synergies and the corresponding R2.

E. Robustness of Task-Specific Synergies

For each movement performed by each subject at each muscle contraction level, a set of muscle synergies were extracted from corresponding MAP matrix using NMF/ANLS algorithm. Robust synergy structure for a certain movement, or the task specificity associated with muscle synergies [39], is important in this study since it increases the potential for reliable movement classification. To this end, we evaluated the robustness of estimated task-specific synergies in the presence of three self-defined force levels for each subject.

1) Variation of Synergy Number: Synergy number has a direct influence on the constitution of estimated muscle syner- gies, which stands a good chance of affecting the performance of synergy-based classification scheme proposed in Section II-G. In view of this, the optimal number of task-specific synergies in the presence of variant force levels was investi- gated herein for each able-bodied subject. And on top of that, we sought to find a generalized synergy number that would work for MSA under all tested conditions.

2) Robustness of Muscle Synergies: It is well known that the MAP involved in a certain task are directly affected by the variation of muscle contraction levels. Therefore, it is necessary to investigate the robustness of estimated synergies against the force variation. We assumed that changing the muscle contraction level within a specific movement would not significantly affect the weighting balance but only the modulus of each synergy vector. Averaged cosine of the angle (ACA) between each pair of synergy vectors was calculated as a criterion to measure their similarity.

$$ACA = \frac{1}{k} \times \sum_{i=1}^{k} \frac{w_{A,i} \cdot w_{B,i}}{\|w_{A,i}\| \, \|w_{B,i}\|}, \quad i = 1, 2, \ldots, k \quad (5)$$

where $\cdot$ denotes dot product, $w_{A,i}$ and $w_{B,i}$ is the i-th synergy vector of task-specific synergies estimated under force level A and B respectively. The scalar result of Equation (5) varies from zero to one, with one denoting the highest similarity. In this study, a scalar value above 0.8 represented high similarity between two muscle synergies because comparable classification results with benchmark methods were observed under this setting.

It should be noted that the synergy vectors in two compared muscle synergies may not perfectly match in order, this is due to the indetermination of NMF/ANLS algorithm. In light of this, we used a best matching algorithm to align the synergy pairs from two estimated synergy matrix. Specifically, the cosine similarity was calculated for all possible combination of synergy pairs between two estimated synergy matrices. The pair with the highest similarity was deemed a match in this study, and the process repeated until all synergy vectors in question were well aligned.

F. Predictive Power of Task-Specific Synergies

The predictive power of task-specific synergies is defined here to describe the reconstruction of original data asso- ciated with a certain task by using muscle synergies esti- mated from the same task. That is, the task-specific synergy matrix W associated with one movement under one force level was remained constant to estimate the activation coefficients matrix H from MAP matrix M within the same movement under another force level:

$$H = W^{+}M \quad (6)$$

where W^{+} is the pseudo-inverse of estimated task-specific synergies. The results of Section III-A indicated that the modulus of synergy vectors scaled with the variation of muscle contraction levels. To mitigate the influence of force variation on estimated muscle synergies, the synergy matrix W were normalized across channels before putting into Equation (6).

Then the derived activation coefficients matrix H were multiplied with the previously estimated task-specific synergies W to reconstruct the corresponding MAP matrix M. The goodness of reconstruction was measured by the VAF criterion. For comparison, task-specific synergies associated with a certain movement were also used to reconstruct MAP matrix of other different movements. Reported for each movement performed by each subject at each force level was the goodness of reconstruction measured by the VAF.

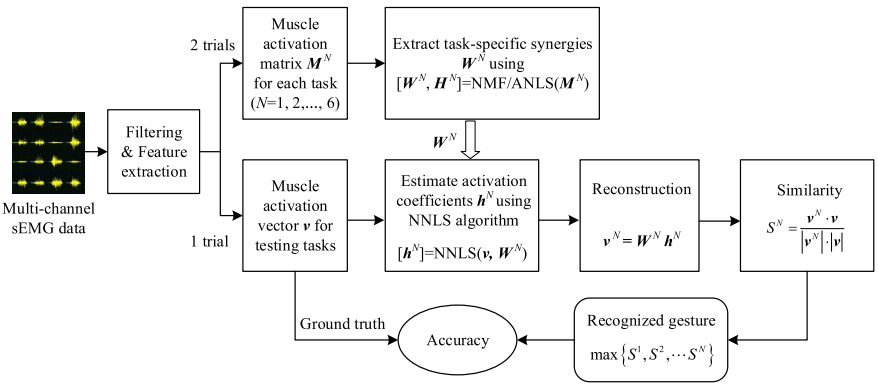


Fig. 1. Synergy-based classification scheme for movement recognition.

G. Movements Identification Based on Muscle Synergy Analysis

An overview of the proposed hand gesture recognition method based on muscle synergy analysis was illustrated in Fig. 1. The assumption here was that the task-specific synergies tended to reconstruct the sEMG data within the same hand gesture better than data associated with other different hand gestures. This assumption was verified in Section III-B.

The whole scheme mainly consisted of two stages: In the training stage, the task-specific synergies W N (N = 1, 2, . . . , 6) for each hand gesture was extracted from cor- responding MAP matrix MN using NMF/ANLS algorithm. These synergy matrixes were then provided one by one to reconstruct the testing data in the testing stage. For each MAP vector v extracted from the analysis window of testing data, the neural commands hN corresponding to each previously learned task-specific synergies were derived by using non-negative least square (NNLS) [40] method:

$$\text{minimize } \frac{1}{2} \| v - W_N h_N \| \quad \text{subject to } h_N \geq 0 \quad (7)$$

To evaluate which task-specific synergies reconstructed the testing samples best, we used the cosine similarity to measure the goodness of reconstruction between the reconstructed MAP vector vN and the original MAP vector v of testing data

$$S_N = v_N \cdot v \, v_N \cdot |v| \quad (8)$$

Task-specific synergies that yielded the highest goodness of reconstruction were picked out and the corresponding move-ment was determined as the unknown movement for testing data. Hereafter, the proposed classification method based on muscle synergy analysis will be abbreviated as MSC.

H. Offline Performance Evaluation

In order to evaluate the effect of variant force levels on the generalizability of proposed classification scheme, three different evaluation scenarios were designed in this study:

1) Intra-force-level evaluation, namely training and testing the algorithm with sEMG data recorded under the same single force level, which is a regular practice in previ- ous studies, however, is too ideal from the real-world conditions.

2) Inter-force-level evaluation, namely training and testing the algorithm with sEMG data recorded under different force levels, which is more approximate to the real-world conditions. 3) Multi-force-level evaluation, namely training the algo- rithm with all three force levels and testing it with single force level data. The offline evaluation was performed through a three-fold cross validation method using sEMG data collected from all three trials for each subject. Namely, data from two out of three trials were used for the estimation of task-specific synergies (or as training data for LDA classifier), and data from the remaining trial was used as testing data. We used the overall classification accuracy (OCA) to measure the classification performance, which was defined as the percentage of correctly identified movements in terms of the total number of testing movements

$$\text{OCA} = \frac{\text{No. of correctly classified movements}}{\text{No. of all testing movements}} \times 100\% \quad (9)$$

For each subject, the algorithm was repeated three times under each force condition so that each trial of the data was fully utilized.

The performance of proposed MSC method was compared with conventional PR-based classification scheme. It has been demonstrated that feature extraction (FE) technique based on discrete Fourier transform and root (DFTR) and time-dependent power spectrum descriptors (TDPSD) both achieved good performance against force variation when combined with linear discrimination analysis (LDA) classifier [12], [13]. We thus chose these two representative methods as the bench-mark for conventional PR system. We also evaluated the effect of different FE techniques on classification accuracy. Specifi-cally, the MSC method was evaluated using RMS, MAV, and DFTR, respectively. Whereas the LDA method was evaluated using DFTR, TDPSD, and TD feature set consisting of MAV, WL, zero crossings (ZC), and slope sign changes (SSC) [41]. Overall, six classification schemes, namely MSC + RMS, MSC + MAV, MSC + DFTR, LDA + TD, LDA + DFTR, LDA + TDPSD were considered in this study.

One-way analysis of variance (ANOVA) was utilized to compare the performance of different classification schemes. Separate ANOVA test was also applied on the MSC method to validate the effect of different training strategies on achieved

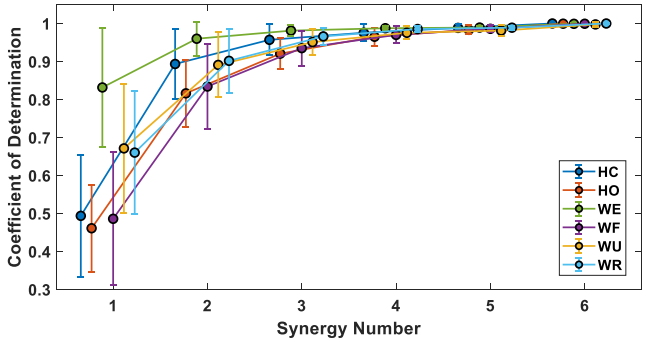


Fig. 2. The R2 versus k plot for each hand gesture. The results were mean and standard deviation of global R2 across all able-bodied subjects and contraction levels.

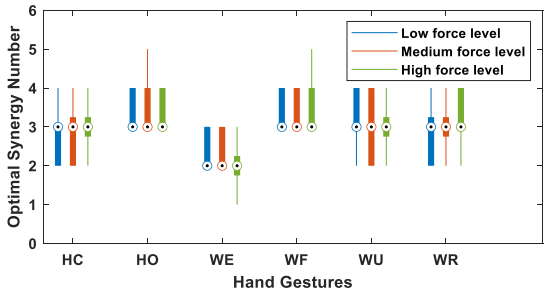


Fig. 3. Box plot for the optimal number of task-specific synergies at three different muscle contraction levels. On each box, the central mark indicates the median, and the bottom and top edges of the box indicate the 25th and 75th percentiles, respectively. The whiskers extend to the most extreme data points. The results were derived from all able-bodied subjects.

classification accuracy. The significance level was set to 0.05 for all tests. The post-hoc analysis with Bonferroni correction method for multiple comparisons was performed when significant difference was detected.

III. RESULTS

A. Robustness of Task-Specific Synergies

1) Variation of Synergy Number: The mean and standard deviation of R2 across all able-bodied subjects as functions of synergy number for each hand gesture were shown in Fig. 2. Overall, the global R2 showed an upward trend with the synergy number increasing from one to six. More specifically, a significant improvement on data reconstruction was observed with a certain synergy number (the optimal synergy number) for each movement, but after that, any further increase in the synergy number only yielded limited improvement on the value of R2.

Fig. 3 illustrated the optimal number of task-specific synergies across all able-bodied subjects at three muscle contraction levels. The box plot indicated that the median of the optimal synergy number ranged from 2 to 3, which means two to three muscle synergies are sufficient to represent the corresponding sEMG data in a low dimensional space. For the convenience of analysis, the optimal synergy number was determined as three for able-bodied subjects and five for amputee subjects in view of different channel number for corresponding EMG data.

2) Robustness of Task-Specific Synergies: Fig. 4 illustrated representative task-specific synergies for each hand gesture

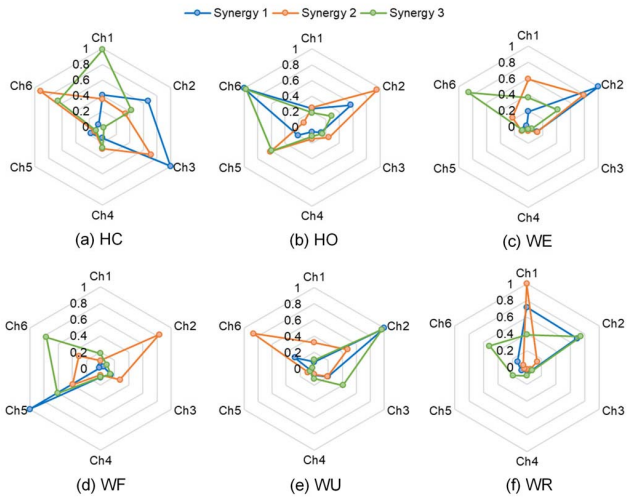


Fig. 4. Representative muscle synergies for each hand gesture performed by one able-bodied subject at medium contraction level. Each color of the radar plot represented one out of three task-specific synergies. The values were normalized to the maximum for each synergy.

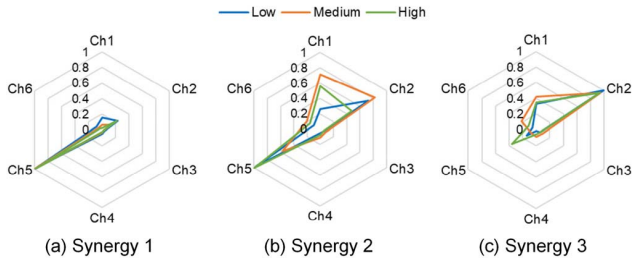


Fig. 5. A set of three task-specific synergies for WR movement performed by one able-bodied subject. Each color represented a muscle contraction level. The values were normalized to the maximum for each synergy.

performed by one able-bodied subject at medium contraction level. A set of three muscle synergies were extracted from the corresponding MAP matrix for each movement. The scaling of each axis in the radar plot represented the weighting of corresponding muscles under the specific EMG channel while performing a certain movement. The weighting values were normalized to the maximum for each synergy. It can be observed that the patterns for each set of task-specific synergies were distinct from one movement to another. The similar structural differences held true for all task-specific synergies extracted from other subjects at other muscle contraction levels.

The set of three task-specific synergies associated with WR movement performed by one able-bodied subject in the setting of three muscle contraction levels were shown in Fig. 5. It can be observed that though the weighting amplitude of each muscle synergy scaled with the increase of muscle contraction level, they were quite similar in terms of the weighting balance (the global patterns in the radar chart). It meant that the structure of each synergy almost remained stable regardless of variant force levels. It was still the case for muscle synergies extracted from other movements performed by other subjects.

The ACA criteria between estimated task-specific synergies while performing each hand gesture under three possible force variation conditions was shown in Fig. 6. The results were derived from all able-bodied subjects. We can see clearly that

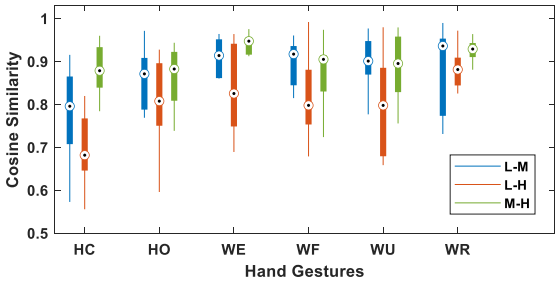


Fig. 6. Box plot for quantitative evaluation of synergy similarity for each movement under three force variation conditions. The results were derived from all able-bodied subjects.

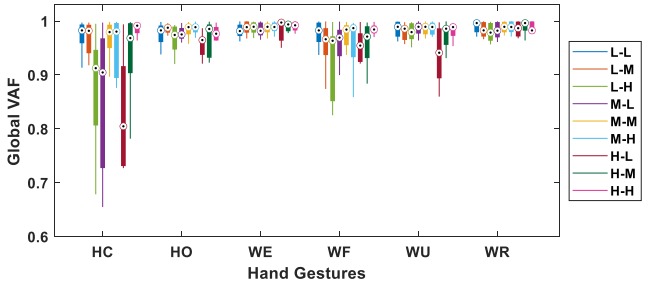


Fig. 7. Box plot for intra-movement predictive power of task-specific synergies for each movement under nine force variation conditions. Each box represents the VAF calculated by using muscle synergies extracted at one force level to reconstruct sEMG data at another force level within the same movement. The results were derived from all able-bodied subjects.

the muscle synergies with small force variation (L-M and M-H) showed high similarity (more than 0.8) for all prede- fined hand gestures. In contrast, low similarity was observed between muscle synergies with large force variation (L-H), especially for muscle synergies associated with HC movement (0.68). Overall, the average similarity across all hand gestures and all force levels was 0.85, higher than the threshold.

B. Predictive Power of Task-Specific Synergies

Fig. 7 illustrated the predictive power of task-specific synergies for the reconstruction of intra-movement data under nine force variation conditions. Good predictive power was observed for muscle synergies associated with each predefined movement, the average VAF across all force variation conditions was 0.92, 0.97, 0.98, 0.95, 0.97, and 0.98, respectively. Looking into details, task-specific synergies showed better predictive power under intra-force-level conditions (with an average VAF of 0.98 across all movements for L-L, M-M, and H-H) than inter-force-level conditions (with an average VAF of 0.96 across all movements for L-M, L-H, M-L, M-H, H-L, and H-M). The predictive power continued to fall for intra-movement data with large force variations. For example, the average VAF declined to 0.92 and 0.95 for L-H and H-L conditions. Moreover, the muscle synergies extracted under medium and high force level showed better predictive power (with an average VAF of 0.98 and 0.97) for the reconstruction of data under other force levels. In contrast, a slight decline in predictive power (with an average VAF of 0.95) was observed for muscle synergies extracted at low force level.

Fig. 8 illustrated the predictive power of task-specific synergies associated with HC movement for the reconstruction of

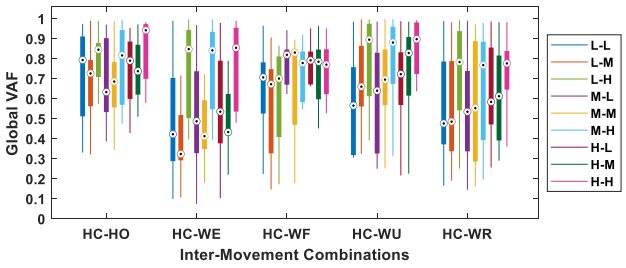


Fig. 8. Box plot for inter-movement predictive power of task-specific synergies under nine force variation conditions. Each box represents the VAF calculated by using muscle synergies extracted from one movement at one force level to reconstruct sEMG data from a different movement at another force level. The results were derived from all able-bodied subjects.

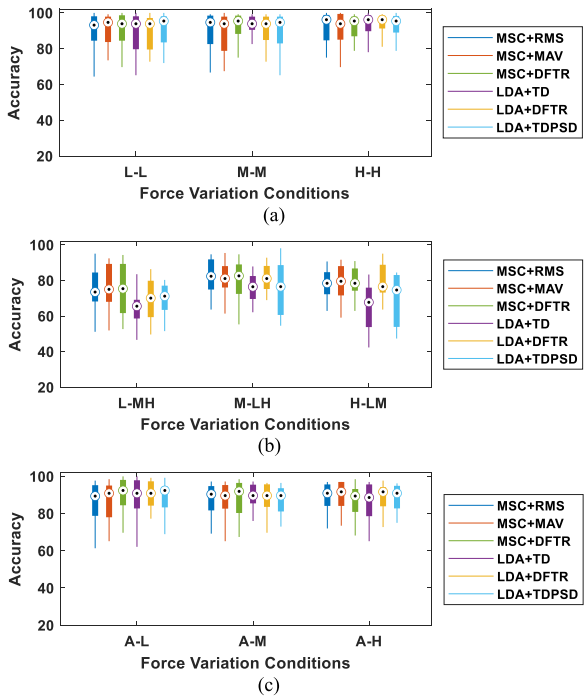


Fig. 9. Box plot for performance evaluation of six classification schemes under nine force variation conditions. (a) Training and testing with the same single force level. (b) Training and testing with different force levels. (c) Training with all three force levels and testing with single force level. The results were derived from all able-bodied subjects.

data from other movements under nine force variation conditions. The average VAF across all force variation conditions was 0.74, 0.59, 0.67, 0.71, and 0.62 for each inter-movement combination. Apparently, there occurred a significant drop in the predictive power for inter-movement data when compared to the intra-movement results shown in Fig. 7. Such deterioration in the predictive power held true for muscle synergies associated with other predefined movements. These results supported the hypothesis proposed in Section II-G that task-specific synergies tended to better reconstruct the sEMG data within the same movement than those from other different movements.

C. Offline Evaluation of Synergy-Based Algorithm

1) Experiments on Able-Bodied Dataset: The performance of six classification schemes was first evaluated on the able-bodied dataset. Fig. 9 illustrated the classification

accuracy achieved in three evaluation scenarios, with an average overall accuracy of 85.2%, 85.3%, 85.8%, 82.5%, 85.6%, and 83.4% for MSC + RMS, MSC + MAV, MSC + DFTR, LDA + TD, LDA + DFTR, and LDA + TDPSD, respectively. In intra-force-level scenario, all classification schemes showed high accuracy when training and testing with sEMG data acquired within the same single force level. The average accuracy in this case (L-L, M-M, and H-H) were 90.2%, 89.4%, 91.3%, 89.5%, 91.4%, and 91.2% for each classification scheme. No significant difference was observed between the performance of six classification schemes ($p = 0.353$).

When it comes to inter-force-level evaluation scenario, namely training and testing with sEMG data collected at different force levels, the accuracy saw a significant decline for all classification schemes. The average accuracy in this case (L-MH, M-LH, and H-LM) were 78.1%, 78.7%, 78.1%, 68.4%, 76.9%, and 71.2% for each classification scheme. Statistical tests revealed significant difference in accuracy between six classification schemes ($p = 0.001$). The pair- wise post-hoc analysis showed that the differences between three synergy-based classification schemes were statistically insignificant ($p > 0.05$). However, the performance of synergy-based method, such as MSC + DFTR, was signifi- cantly higher than LDA + TD scheme (mean difference: 9.7%, $p = 0.027$) and LDA + TDPSD scheme (mean difference: 6.9%, $p = 0.041$). It is worth noting that the synergy-based method achieved higher accuracy when training at medium/high force level than that training at low force level. The average accuracy of MSC + DFTR when training at low, medium and high force level was 82.4%, 87.9%, and 85.8%, respectively. This result was in line with the outcome of the predictive power of task-specific synergies in Section III-B. Statistical test revealed significant effects of training strategy on the generalization accuracy of MSC + DFTR method ($p = 0.027$). A pair-wise post-hoc analysis indicated the highest accuracy was achieved when training at medium force level (medium-low difference: 5.5%, $p = 0.033$). No significant difference was observed when training at medium and high force level ($p > 0.05$).

In multi-force-level evaluation scenario, better performance was observed for all classification schemes as compared to inter-force-level scenario, but the overall accuracy in this case was not as high as that in intra-force-level scenario. The average accuracy across three force variation conditions (A-L, A-M, A-H) were 87.3%, 87.7%, 88.2%, 86.9%, 88.6%, and 88.1% for each classification scheme. Statistical tests showed no significant difference between the performance of six classification schemes in this case ($p = 0.695$).

2) Experiments on Amputee Dataset: We further evaluated the performance of different classification schemes on the amputee dataset. Fig. 10 depicted the classification accuracy achieved under nine force variation conditions for MSC + DFTR, LDA + DFTR and LDA + TDPSD, as the superiority of these methods have been demonstrated in Fig. 9 and in previous literature. Obviously, the performance on amputee subjects was not as good as that on able-bodied subjects for all classification schemes investigated. Specifically, the

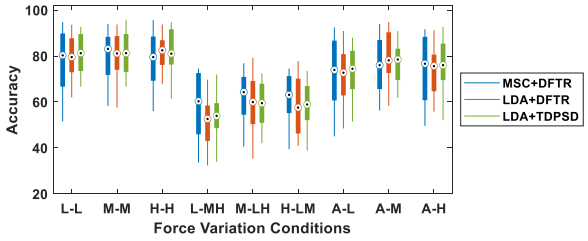


Fig. 10. Box plot for performance evaluation of representative classifi- cation schemes under nine force variation conditions. The results were derived from all amputee subjects.

TABLE I COMPUTATIONAL EFFICIENCY FOR SYNERGY-BASED CLASSIFICATION SCHEME WITH REGARD TO DIFFERENT FEATURE EXTRACTION TECHNIQUES

Features	Time for FE (ms)	Time for GC (ms)	Controller Delay (ms)
RMS	0.039	1.364	113.903
MAV	0.027	1.364	113.891
TD	0.254	1.364	114.118
DFTR	0.432	1.364	114.296

average accuracy of MSC + DFTR dropped by 13.1% and 13.9% for intra-force-level scenario (L-L, M-M, and H-H) and multi-force-level scenario (A-L, A-M, and A-H) respec- tively. As for the inter-force-level scenario (L-MH, M-LH, and H-LM), the average accuracy dropped even further by 17.4%. Statistical tests showed significant differences between MSC + DFTR and other classification schemes in inter-force- level scenario, with $p < 0.001$ for MSC + DFTR versus LDA + DFTR, $p = 0.0157$ for MSC + DFTR versus LDA + TDPSD. No significant difference was observed in intra-force-level scenario ($p = 0.619$) and multi-force-level scenario ($p = 0.733$). Overall, the average accuracy across all nine force variation conditions were 71.2%, 70.5%, and 71.5% for each classification scheme.

3) Computational Efficiency: Finally, we investigated the computational efficiency of proposed MSC method to evaluate its potential for clinically viable myoelectric control. The experiment was performed using MATLAB 2016b on a computer with 3.6 GHz Intel Core i7 processor, 16GB RAM. TABLE I listed the average computational efficiency associated with feature extraction (FE) and gesture classification (GC), from overlapped analysis window of 200 ms length and 25 ms increment. The time required for gesture classifica- tion (GC) is 1.364 ms for MSC method. The controller delay for proposed scheme is therefore within 120 ms according to [42], which is applicable for myoelectric controlled pros- thesis [43].

IV. DISCUSSION

Previous studies either independently demonstrated the existence of MS in human motor control, or simply integrated MS into movement classification under equal muscle contraction level. However, no prior study has concurrently investigated the effect of force variation on both the synergy structure and synergy-based classification performance. The main strength of present study was that it gave an experimental insight into

the reliability of MS in the setting of variant force levels. We also proposed a synergy-based classification method against force variation, which is a serious handicap in real- world myoelectric prostheses control. The proposed method is very different from conventional PR-based method that uses different features to train a classifier for task identification. Instead, we extract task-specific synergies from multichannel EMG and utilize their predictive power for task identification. Ultimately, we want to provide an alternative approach for the development of clinically viable prostheses and rehabilitation robots powered by sEMG.

We first investigated the underlying physiological informa- tion hidden in acquired sEMG signals by means of muscle synergy analysis. Task-specific synergies extracted from six predefined hand gestures showed significant structural differ- ences (Fig. 4) due to synergy specificity previously described in [39]. In contrast, high similarity (scaling amplitude with almost fix weighting balance) was observed among synergies extracted within the same movement in the presence of variant force levels (Fig. 5). This is in line with the results of [32] that muscle synergies are relatively fixed across variant con- straints. Large force variation was more likely to change the synergy structure since the similarity between L-H force level were lower than the similarity between L-M force level and M-H force level (Fig. 6). In addition, task-specific synergies showed better predictive power for intra-movement data than for inter-movement data (Fig. 7 and Fig. 8). These outcomes are the cornerstone for the proposed movement classification method. We noted that a similar classification method was introduced in [16] and the reported performance was better than ours. However, they utilized a post-processing procedure to improve the accuracy and their experimental scheme was constrained within the same force level, which is too ideal for the real-world conditions. In contrast, we extended the constraints into variant force levels and employed no post-processing procedure. Besides, we chose the NMF/ANLS algorithm and NNLS algorithm to estimate muscle syner- gies and neural commands respectively, which is more cost-effective and the results are of greater physical importance. To our best knowledge, this was the first time to concurrently demonstrate the robustness and predictive power of task-specific synergies under variant force levels and explore their application in reliable movement classification against force variation.

We compared the performance of synergy-based method with conventional PR-based method consisting of LDA clas- sifier and representative FE techniques reported in literature. The classification accuracy saw a significant decline for all classification schemes when sEMG data used for training and testing were from different force levels (Fig. 9). This result agrees with the results of previous studies that force variation may degrade the performance of myoelectric classification [3]. Statistical tests indicated better performance of proposed method against force variation than conventional PR-based method. This may suggest potential application of synergy-based classification method into clinically viable prosthetics against force variation.

The proposed method showed relatively higher accuracy when trained at medium and high force levels than trained at low force level (Fig. 9). It means that reasonable train- ing strategy with data acquired at a single force level (medium or high) can significantly enhance the generaliza- tion of MSC method in the setting of variant force levels. Although the accuracy improvement in this case was less than that training with multiple force levels (Fig. 9), this synergy-based classification scheme may largely reduce the training time for potential application in upper-limb prosthe- ses with intuitive and robust myoelectric control. Moreover, the processing time required for the estimation of neural commands and movement classification is approximately 1.364 ms, which shows good prospect for real time prosthesis control.

The limitation of this study lies within two aspects: Firstly, the present method was focused on steady-state portion of EMG data with limited transient portion between tasks. How- ever, this might not be a significant problem since the effect of transient information on the classification performance could be minimized by using a threshold to detect the onset of the motion [44]. Secondly, the proposed method was only evaluated through offline analysis since the main objective of this study was to check the utility of synergy-based method against force variation. We thus suppose that offline analysis is sufficient to reach the goal at the moment. In fact, the construction of an online system for the control of a prosthetic hand is underway and the detailed results of these trials will be reported in future publications.

V. CONCLUSION

In this paper, we investigated the robustness and predictive power of task-specific synergies under three self-defined mus- cle contraction levels, and presented a reliable classification scheme against force variation. Muscle synergy analysis of 6-channel sEMG signals showed that a three-synergy model associated with each of six predefined hand gestures was able to explain at least 90% of the variance in corresponding MAP matrix; The weighting balance of each task-specific synergies remained stable regardless of force variation, though their weighting amplitude scaled with the increase of muscle con- traction level. These results support muscle synergy analysis as a robust tool to interpret the neurophysiological information and motor intent hidden in sEMG signals; Statistical tests indicated better performance of proposed method against force variation than conventional PR-based method. We recom- mended a reasonable training strategy for satisfactory accuracy by training the scheme at medium or high force level. The controller delay of MSC method lies within 120 ms owing to the NMF/ANLS algorithm and NNLS algorithm used in the estimation of muscle synergies and neural commands respectively. These outcomes throw light on our future work towards clinically reliable myoelectric control of upper-limb prostheses. It also provides an alternative method for the development of more intuitive and natural human machine interfaces in rehabilitation robots and so on.

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